Machine Learning for Mechanical Engineers at CoEP Session 9: Logistic Regression

Session 9: LogiRegr

Introduction

Answering yes/no questions

Often we have to resolve questions with binary or yes/no outcomes.

- Does a patient have cancer?
- Will a team win the next game?
- Will the customer buy my product?
- Will I get the loan?

Continuous vs Categorical Variables

Types of variables:

- Continuous: age, income, height : use numerical (float) value
- Categorical (discrete): gender, city, ethnicity : use enums (ints to an extent)

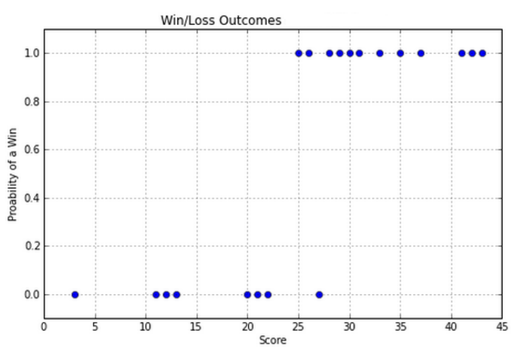
Example: Win/Lose based on Scores

Start by plotting variables that decide the outcome on x axis and the outcome on the y axis.

- X-axis is the number of points scored by a team
- Y-axis is whether they lost or won the game (0 or 1)
- General idea: more the score, more chances of winning, and vice versa.

Any guesses on how the plot would look like?

Plot



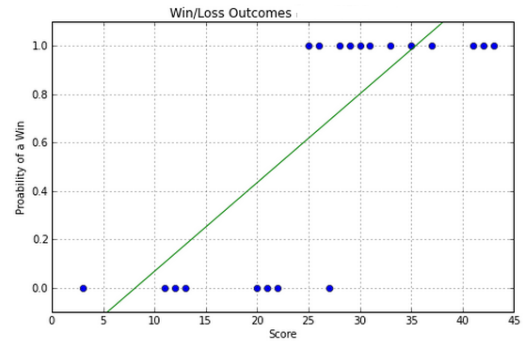
So, how do we predict whether we have a win or a loss if we are given a score?

Game score prediction

- Can you fit a line?
- Sure, you can, but is represent the data?
- What would be a good representation/model?

Game score prediction

Take a look at this plot of a "best fit" line over the points.

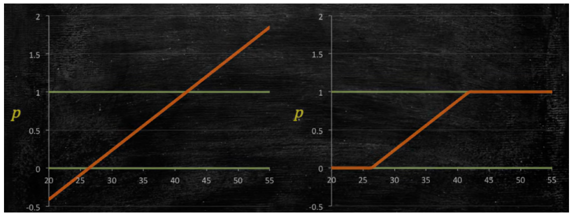


Why not Linear Regression, here?

- Probability of winning is bound, from 0 to 1.
- Scores can range from 0 to 50, say.
- Now we need a function (or a model) that, given a new score, say, 30. it should compute value between 0 to 1. From the graph, it comes out to be 0.8. Its close to 1, so we can predict that the team will win.
- Similarly, for score 10, probability is 0.08, very low, so we can predict that the team will lose.
- But what for scores, 2, or 45. The probabilities are coming out of the range of 0 to 1. Bad!!!

Why not Linear Regression, here?

- Well, we can say, value calculated above 1 is 1, and similarly any value calculated below 0 is considered to be 0. Some sort of Capping.
- Then the function would look like mirrored "Z". It has kinks. Too specific!!, not good.
- So, clearly linear regression is not a good model.
- Can we find a better function?



What do we need here?

- We need some magic function to represent this requirement.
- Input: any number
- Output: either Boolean or even probabilities between 0 to 1 would do. Using threshold we will convert them to Booleans.

Logistic Regression

- For example, consider a logistic regression model for spam detection.
- If the model infers a value of 0.932 on a particular email message, it implies a 93.2% probability that the email message is spam.
- More precisely, it means that in the limit of infinite training examples, the set of examples for which the model predicts 0.932 will actually be spam 93.2% of the time and the remaining 6.8% will not.
- Choice of threshold is an important choice, and can be tuned

Logistic Regression

- The name of the Magic function is *logit*.
- The theory is based on a term called "Odds".
- Lets look at them in details, next.

Logit

- p being probability can be only between 0 to 1.
- Taking log of odds, gives the "logit" function $logit(p) = log_e \frac{p}{(1-p)}$
- Whats the range of logit values?

(Reference: Simple Linear Regression: Step-By-Step - Dan Wellisch)

Range of Logit

- Put p = 0.5, $logit(0.5) = log \frac{0.5}{(0.5)} = 0$
- Put p = 0.1, $logit(0.1) = log \frac{0.1}{(0.9)} = -0.95$
- Put p = 0.01, $logit(0.01) = log \frac{0.01}{(0.99)} = -1.99$
- Put p = 0.9, $logit(0.9) = log \frac{0.9}{(0.1)} = 0.95$
- Put p = 0.99, $logit(0.99) = log \frac{0.99}{(0.01)} = 1.99$
- Put p = 0.999, $logit(0.999) = log \frac{0.999}{(0.001)} = 2.99$

So, the logit function takes input values in the range 0 to 1 and transforms them to values over the entire real number range.

Reverse

We want something inverse ...

- $logit(p) = log_e \frac{p}{(1-p)}, p = 0 \rightarrow 1$
- We are interested in reverse
- $\phi(z) = logit^{-1}(z) = \frac{1}{1+e^{-z}}, z = -\infty \rightarrow \infty$
- z is any number. In our case its a linear combination of variables giving any number, and the logit inverse will return the probability
- $logit^{-1}(z)$ is called as Sigmoid function

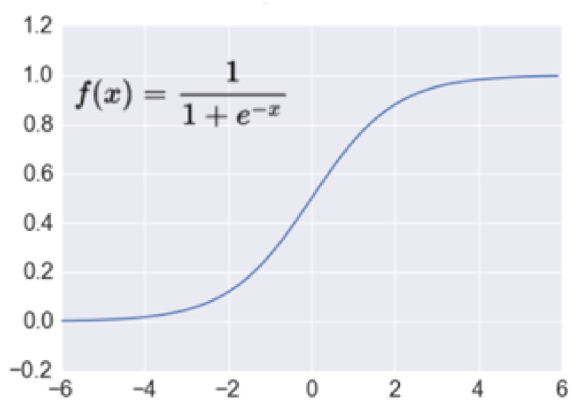
Plotting Sigmoid

```python
import matplotlib.pyplot as plt
import numpy as np

def sigmoid(z):
  return 1.0 / (1.0 + np.exp(-z))

z = np.arange(-7, 7, 0.1)
phi_z = sigmoid(z)
plt.plot(z, phi_z)
plt.axvline(0.0, color='k')
plt.axhspan(0.0, 1.0, facecolor='1.0', alpha=1.0, ls='dotted')
plt.axhline(y=0.5, ls='dotted', color='k')
plt.yticks([0.0, 0.5, 1.0])
plt.ylim(-0.1, 1.1)
plt.xlabel('z')
plt.ylabel('$\phi (z)$')
plt.show()
```

Sigmoid



Sigmoid

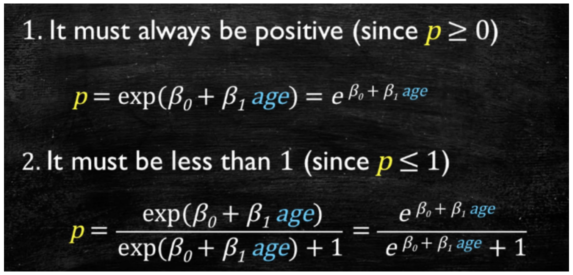
$\phi(z) = logit^{-1}(z) = \frac{1}{1+e^{-z}}$

- If z goes towards infinity then e^{-z} becomes very small , denominator almost 1, thus $\phi(z)$ becomes almost 1.
- If z goes towards minus infinity then e^{-z} becomes very large , denominator almost infinity, thus $\phi(z)$ becomes almost 0.
- Thus, we conclude that this sigmoid function takes real number values as input and transforms them to values in the range [0, 1] with an intercept at $\phi(z) = 0.5$

Logistic Regression Algorithm

The Magic function

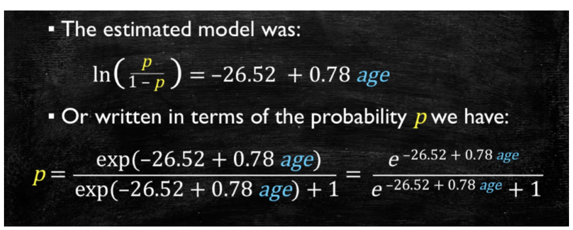
- Example problem, say, that chances of you getting Diabetes is more with age. So, we are finding a function that takes expression $\beta_0 + \beta_1 age$ as input and outputs a probability.
- For probability, 2 conditions are that the value is always positive and less than 1.
- Which functions always output positive value given any (positive or negative) value?
- 'Absolute', 'Square', etc. One more is exponential.
- e^z is always positive whatever the z is. But it can be greater than 1.
- To make it less than 1, lets divide the same term but slightly larger, say, with plus 1. Nice Trick!!



(Ref: Logistic Regression - Interpretation of Coefficients and Forecasting - Data Mining In CAE)

The Magic function

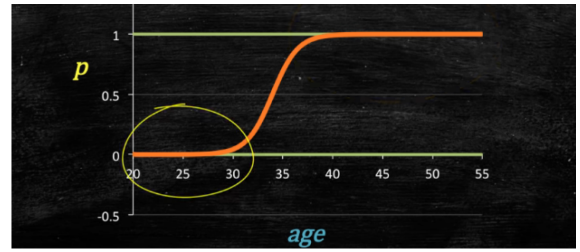
- Rearranging previous equation.
- Lets say $z = \beta_0 + \beta_1 age$. So we have $p = \frac{e^z}{e^z+1}$
- Is it same as $p = \frac{1}{1+e^{-z}}$? Can not believe? Try.
- Arranging p terms and taking log on both sides $ln(\frac{p}{1-p}) = z = \beta_0 + \beta_1 age$
- So even though the probability function is not linear (its exponential), the above arrangement is Linear function of age. Very similar to Linear Regression.



(Ref: Logistic Regression - Interpretation of Coefficients and Forecasting - Data Mining In CAE)

The Magic function

- Plot of Probability vs age looks like a Signmod
- As the age increases, the chances of getting diabetes are almost equal to 1 (Asymptotic).
- As the age lowers, the chances of getting diabetes are almost equal to 0 (Asymptotic).
- $logit(p) = \log(p/(1-p)) = b_0 + b_1x_1 + \ldots + b_kx_k$
- The $ln(\frac{p}{1-p})$ is the logit function and thus the algorithm is called as Logistic.
- It has Regression expression on the right, so its called Regression.
- Thus this is Logistic Regression. Used for Classification



(Ref: Logistic Regression - Interpretation of Coefficients and Forecasting - Data Mining In CAE)

Intuition . . .

- Whats the meaning of Linear expression's coefficients?
- In Linear Regression, it meant slope, or proportionality of increase of y wrt x.
- Here thats not the case.
- Every unit increase in age, $\log(p/(1-p))$ increases β_1 units.
- Say, for age as 35, $\beta_0 = -26.52$ and $\beta_1 = 0.78$, the regression comes to be $= -26.52 + 0.78 \times 35 = 0.813$.
- Is it Probability of having Diabetes??
- No. Its logit value.
- $p = \frac{e^z}{e^z+1}$ where $z = 0.813$.
- Result: Probability $p = 0.693$
- Approx 70% chance.
- Exercise: Calculate for age 36.
- Confirm: $z = 1.594$ and Probability comes to be 0.831. Just a marginal increase in age, far more chances of getting disease. But after certain age, say 45, the probability settles towards 1.

Intuition
Linear regression's straight line has no trouble predicting a probability of -0.3 or 1.8, which is nonsense for something that must stay between 0 and 1. Logistic regression sidesteps this by first fitting a straight line in "logit" (log-odds) space, which is unbounded like any linear expression, and only converting to a probability at the very end via $p = e^z/(e^z+1)$ – a curve that squashes any real number into $(0, 1)$, never quite reaching 0 or 1.

Intuition . . .

- You can add one more variable to the regression part.
- We add Gender.
- $z = \beta_0 + \beta_1 age + \beta_1 woman$
- Assume that betas are calculated and the equation becomes: $z = -26.47 + 0.79 \times age - 0.56 \times woman$
- One year increase in Woman's age increases risk by 14%, and thats 12% for male.

Variable	Coefficients	35y W	35y M	36y W	36 M
Constant	-26.465300	1	1	1	1
Age	0.787213	35	35	36	36
Woman	-0.557795	1	0	1	0
y* = ln(p/(1-p))		0.529	1.087	1.317	1.874
p = exp(y*)/(exp(y*)+1)		0.629	0.748	0.789	0.867

(Ref: Logistic Regression - Interpretation of Coefficients and Forecasting - Data Mining In CAE)

Optimization . . .

- $logit(p) = \log(p/(1-p)) = b_0 + b_1 x_1 + \ldots + b_k x_k$
- Above, p is the probability of presence of the characteristic of interest.
- It chooses parameters that maximize the likelihood of observing the sample values rather than that minimize the sum of squared errors (like in ordinary regression).

Conclusion . . .

For better understanding . . .

- So, by now it is clear that the Logistic Regression is somewhat a misnomer!
- It is a classification not a regression algorithm problems.
- We can imagine that the Logistic Regression is done in two parts.
- In first, using X features, some intermediate y value is predicted, like Regression.
- This value is then passed via this Analog to Digital Converter like Sigmoid function, resulting into 0 or 1.

Evaluation Metrics

Accuracy

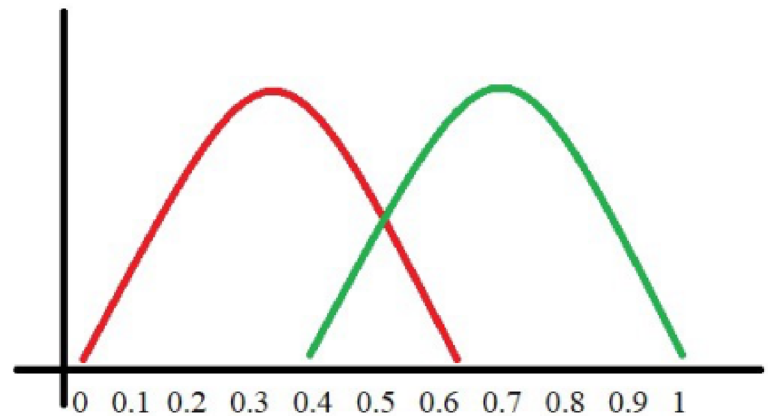
- How do we evaluate classification models?
- One possible measure: Accuracy
  - the fraction of predictions we got right
- Accuracy Can Be Misleading

Accuracy

- In case of cancer dataset, if you blatantly predict "No Cancer" you are going to be right 99% times, as thats the percent of Non-Cancer patients in the population.
- In this case, such great accuracy is actually useless, as the class/target is imbalanced.
- Need Confusion Matrix.

What is ROC?

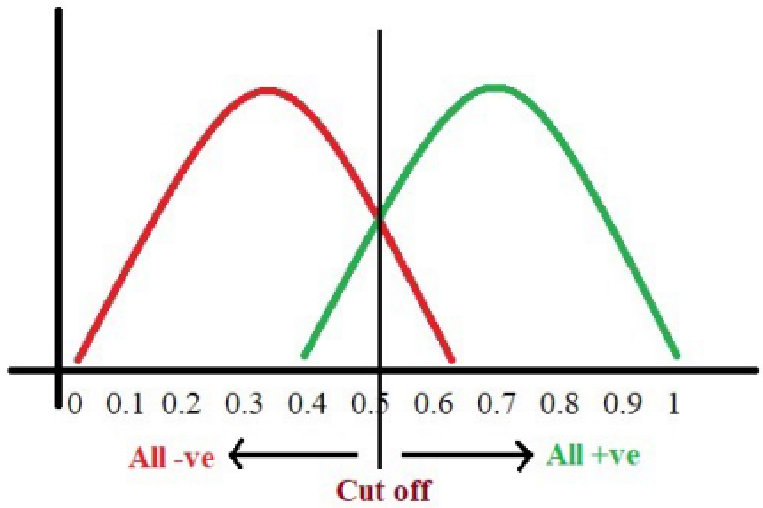
- ROC : Receiver Operating Characteristics Curve
- ROC tells us about how good the model can distinguish between two things (e.g If a patient has a disease or no).
- Better models can accurately distinguish between the two.
- Predictions are plotted on x, and colors show True or false. Y is frequency or counts.

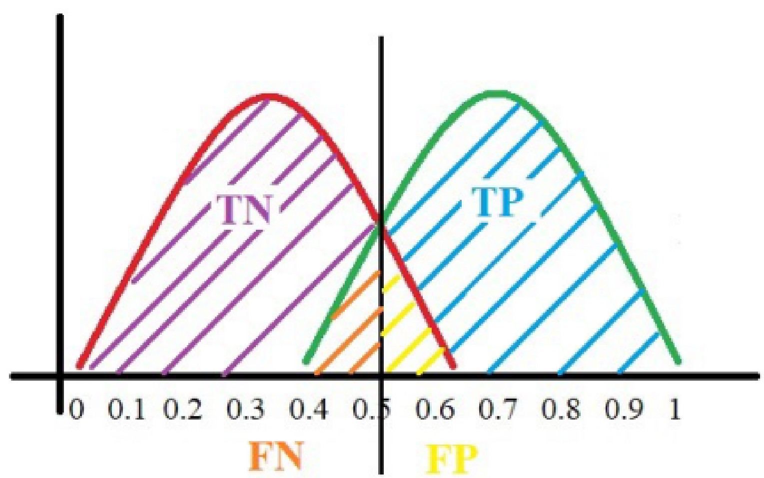


(Ref: Let's learn about AUC ROC Curve! – GreyAtom)

What is ROC?

- Pick a value where we need to set the cut off say, "0.5" as shown
- All the positive values above the threshold will be "True Positives"
- Negative values above the threshold will be "False Positives" as they are predicted incorrectly as positives.
- All the negative values below the threshold will be "True Negatives"
- Positive values below the threshold will be "False Negative" as they are predicted incorrectly as negatives.





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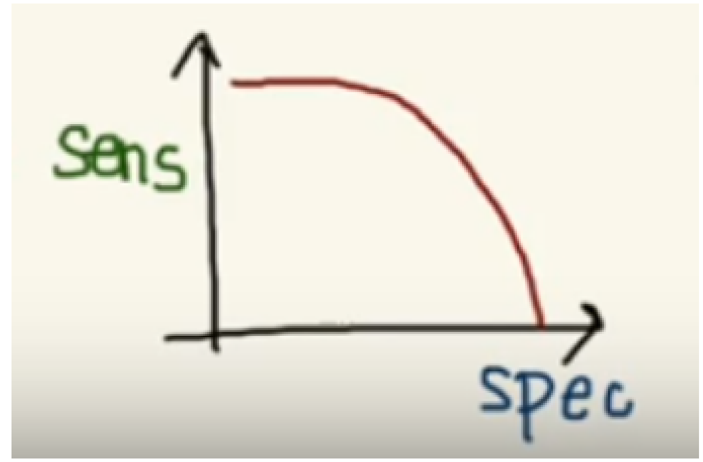
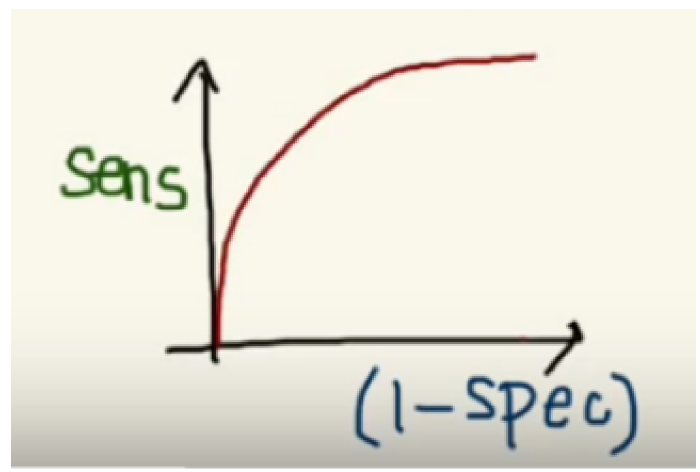
What is Sensitivity and Specificity?

- Sensitivity or Recall = TP/TP + FN.
- Specificity = TN/TN + FP.
- When we decrease the threshold, we get more positive values thus increasing the sensitivity. Meanwhile, this will decrease the specificity.
- and vice versa. Meaning, there is a trade-off.

- Instead of Specificity we use (1 - Specificity) and the graph will look something like the bottom one:



Trade off between Sensitivity & Specificity



(Ref: Let's learn about AUC ROC Curve! – GreyAtom)

Why do we use (1- Specificity)?

- Let's derive what exactly is (1-Specificity)
- Specificity gives us the True Negative Rate
- (1 - Specificity) gives us the False Positive Rate Rate
- So now we are just looking at the positives.
- As we increase the threshold, we decrease the TPR as well as the FPR and when we decrease the threshold, we are increasing the TPR and FPR.

$$Specificity = \frac{TN}{TN + FP}$$

$$1 - Specificity = 1 - \frac{TN}{TN + FP}$$

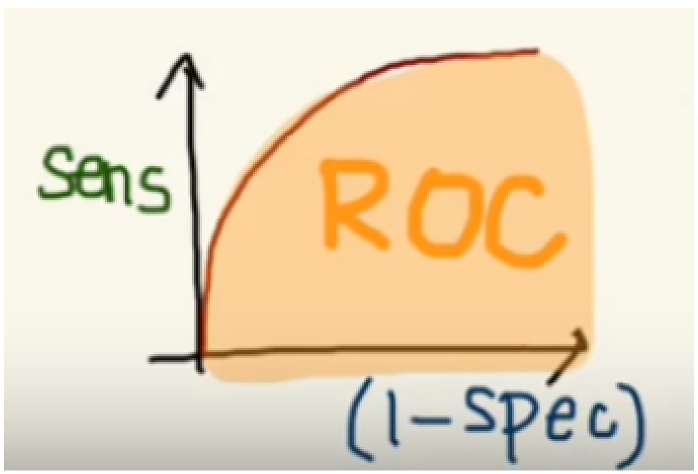
$$1 - Specificity = \frac{TN + FP - TN}{TN + FP}$$

$$1 - Specificity = \frac{FP}{TN + FP}$$

(Ref: Let's learn about AUC ROC Curve! – GreyAtom)

What is AUC?

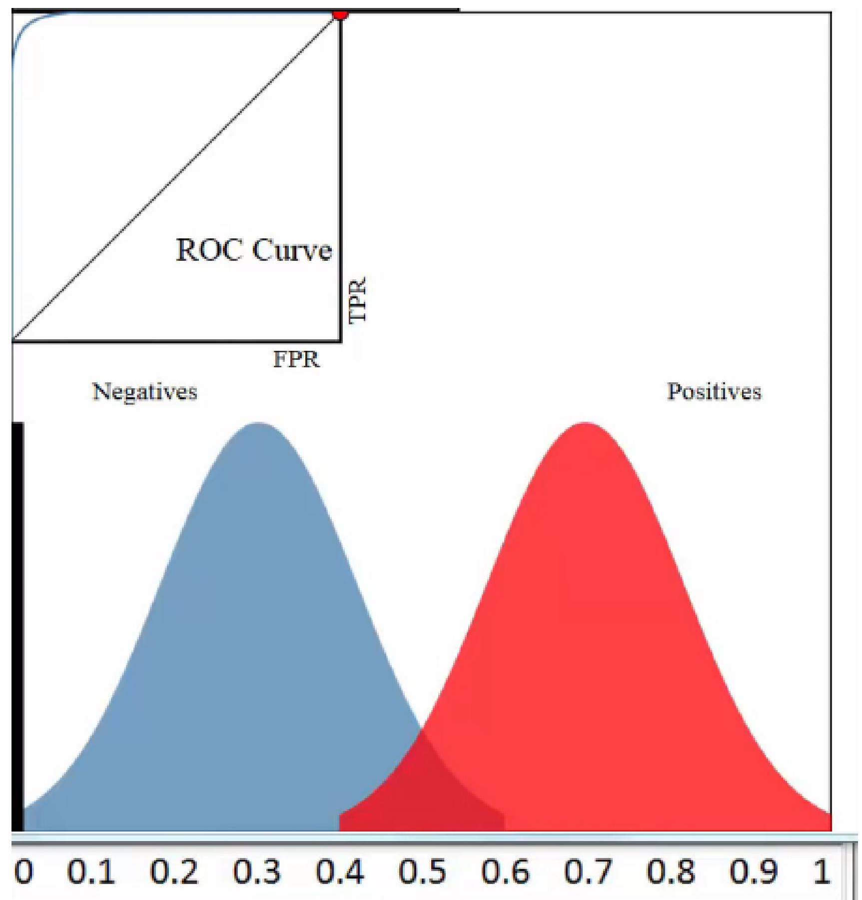
- The AUC is the area under the ROC curve.
- It gives us a good idea of how well the model performances.
- AUC ROC indicates how well the probabilities from the positive classes are separated from the negative classes.



(Ref: Let's learn about AUC ROC Curve! – GreyAtom)

ROC AUC Intuition (More Details)

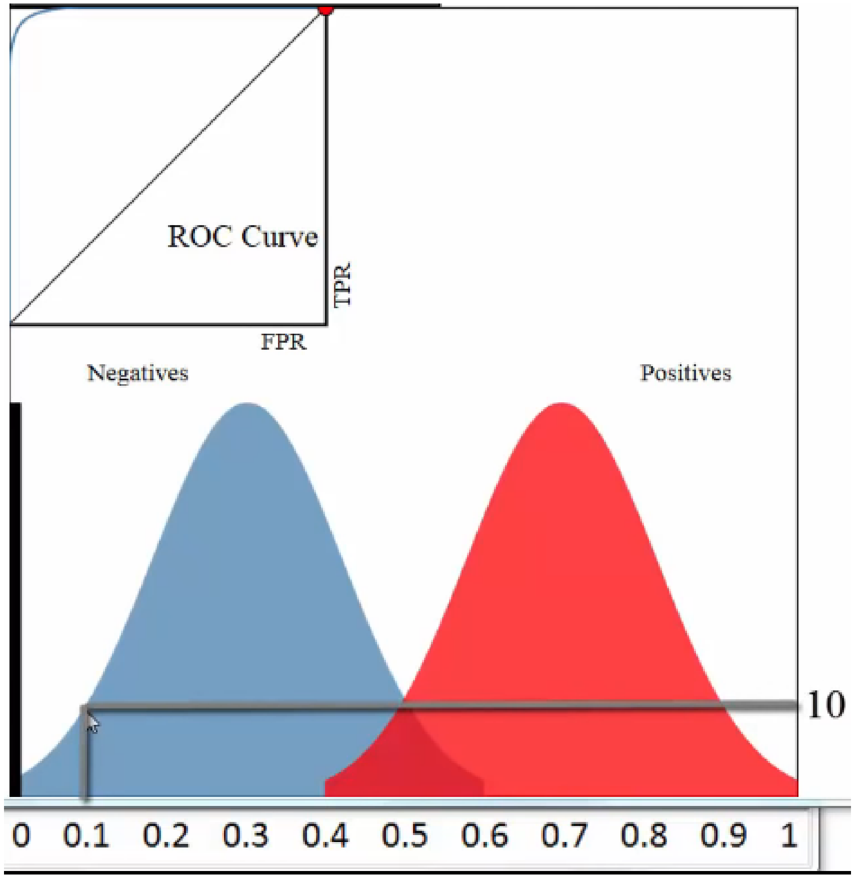
- ROC : Receiver Operating Characteristics Curve
- AUC: Area under curve
- Often used to evaluate performance of a binary classifier
- Two possible outcomes: Positive and Negative
- This is histogram (continuous) where red pixels are positive (got admission to IIT!!) and blue pixels are negative (did not get admitted)
- 250 students (pixels here, for example) got admitted and 250 did not get admitted.



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

- This is result of predictions done on validation set. For which, you have actual values as well.
- The classifier is like Logistic Regression where you get probabilities as well.
- Predicted probabilities are seen on the x axis. Y axis is count of students with ACTUAL red(admitted) or blue (not admitted) values
- E.g. there were 10 students for which you predicted admission probability of 0.1, they did not get admitted shown as negative (blue)

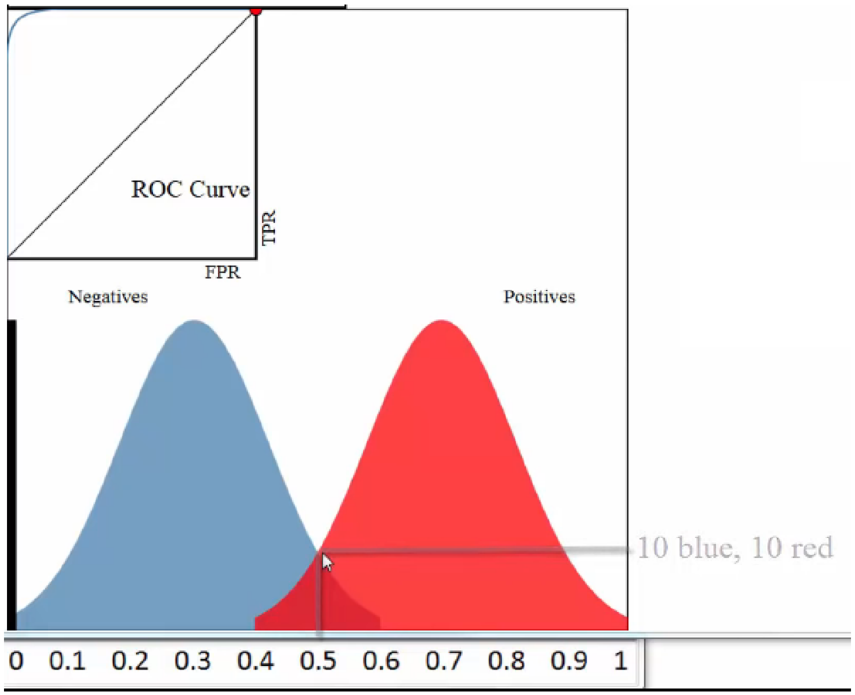


(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

- E.g. there were 50 students for which you predicted admission probability of 0.3, they did not get admitted, ie negative (blue) (not shown here, but height of the bell curve is 50)
- For 20 students where you predicted 0.5 probability of admission, 10 did not get admitted (blue) but 10 got admitted (red)

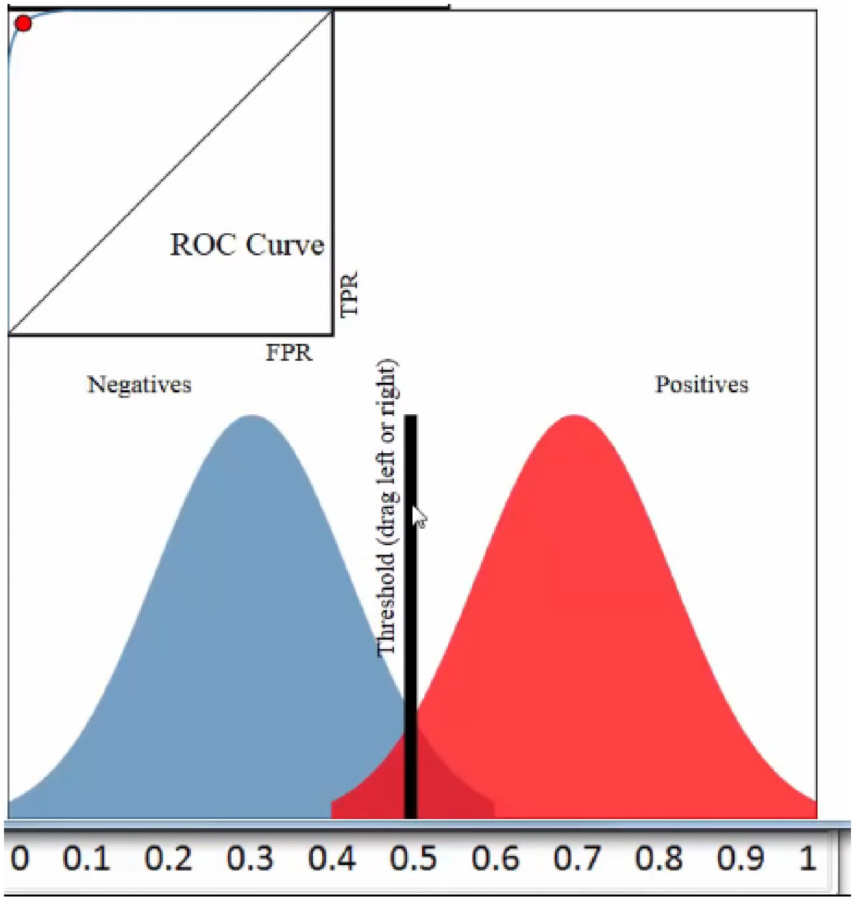
- You can say that the classifier is doing well as it is able to partition/separate two classes reasonably well (only small overlap or ambiguity in the middle)



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

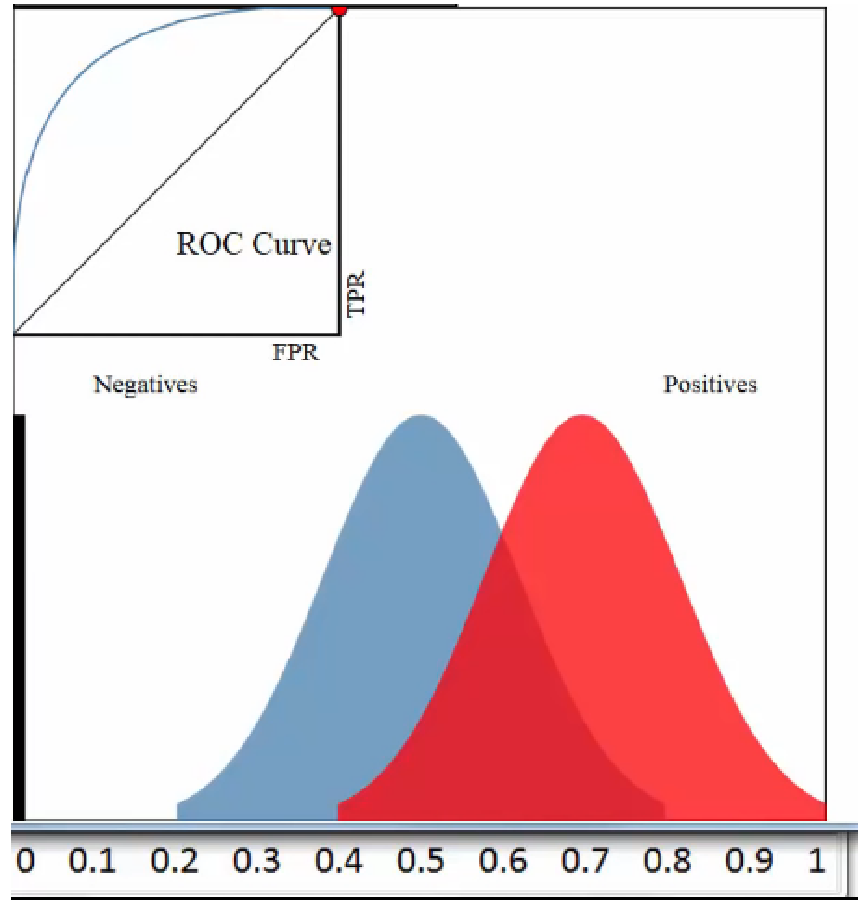
- So. we can put threshold (or the partition) at 0.5, anything about that is admitted, and below is not admitted.
- With this threshold the accuracy looks to be in the range of 90%.
- Please not that the ROC curve in the top left is almost touching the left and top boundaries. Its bit away at the corner due to some errors in the middle region.



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

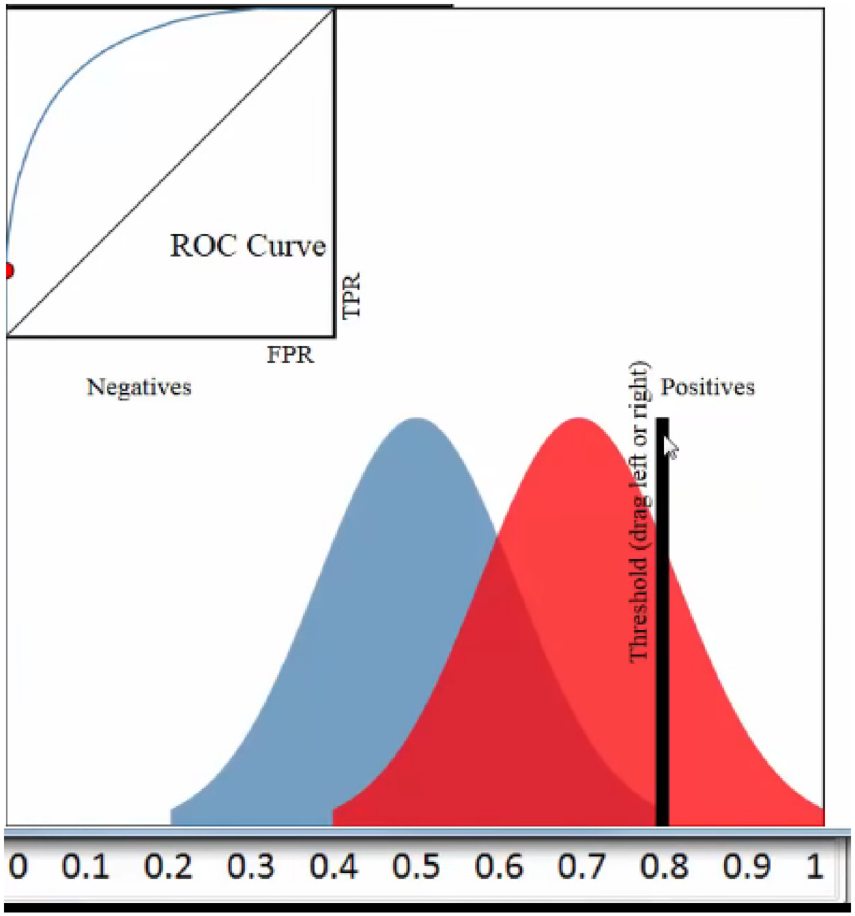
- If the classifier was not good, there would be far more overlap in the middle,ie many misclassifications.
- Regardless of where you put threshold, your classification accuracy will be bad
- Note that the ROC curve has shifted and too much gap way from the top left corner, so many errors.
- ROC curve is a plot of True Positive Rate (TPR) on y axis and False Positive Rate (FPR) on x axis for every possible classification threshold (say, 0 to 1)



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

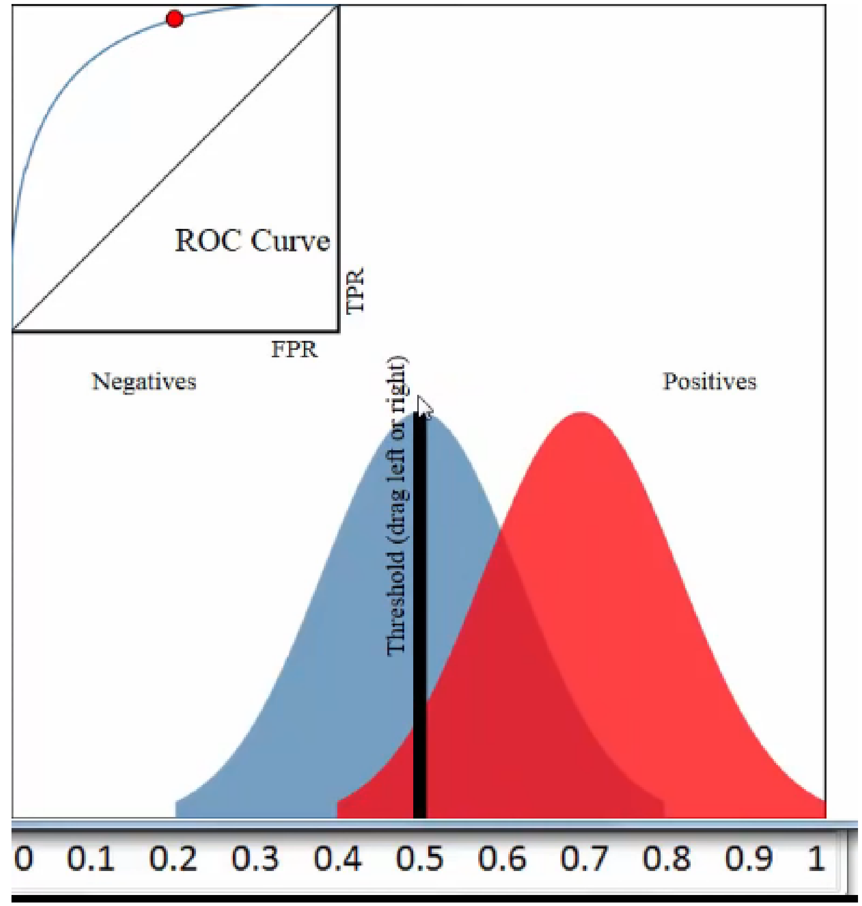
- TPR : True Positives Predicted divided by All actual positives
- FPR : False Positives Predicted divided by All actual negatives
- Both range from 0 to 1.
- For threshold 0.8, TPR is pixels on right (approx 50) divided by all reds (actual positives) ie 250, so $50/250 = 0.2$,
- FPR is blue pixels on right of the line (0) divided by all blues ie 250, so $0/250 = 0$
- Plot point at $x = 0$, and $y = 0.2$



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

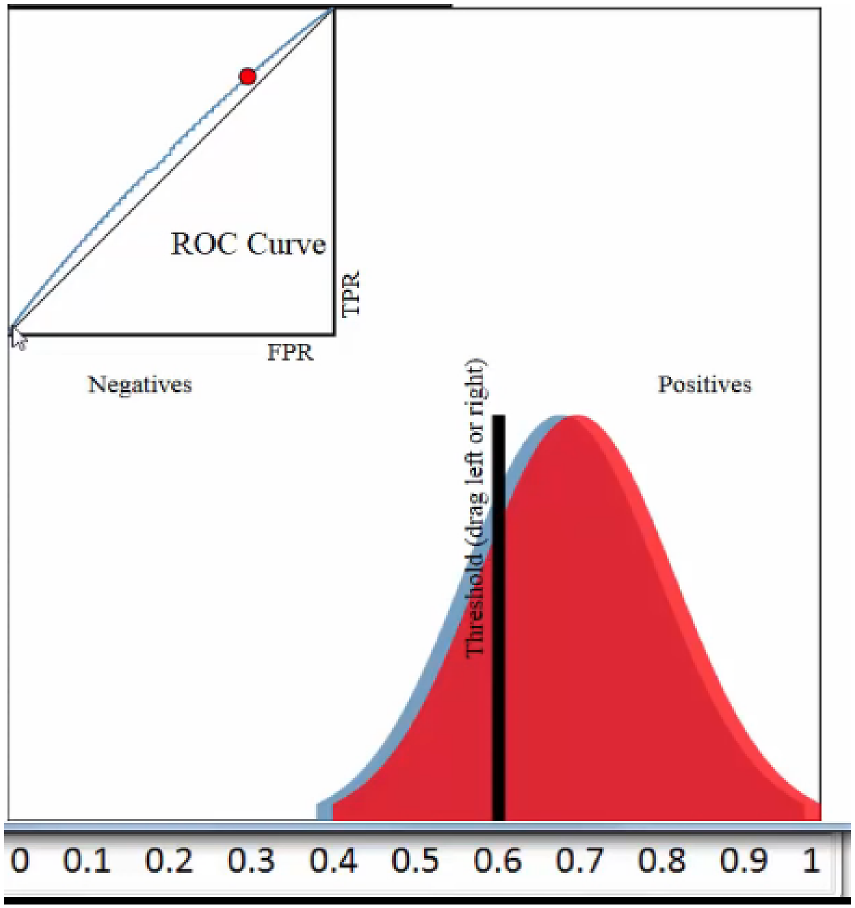
- For threshold 0.5, TPR is pixels on right (approx 230) divided by all reds (actual positives) ie 250, so $230/250 = 0.94$,
- FPR is blue pixels on right of the line (125) divided by all blues ie 250, so $125/250 = 0.5$
- Plot point at $x = 0.5$, and $y = 0.94$



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

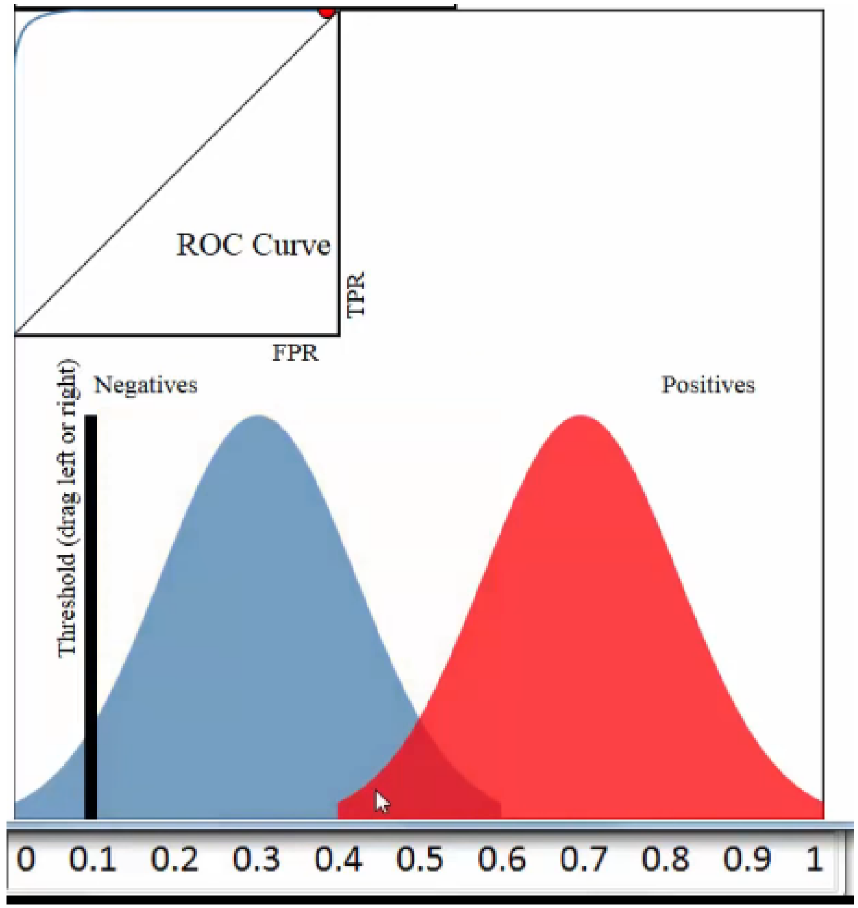
- Similarly plot for all possible thresholds.
- Thus ROC curve visualizes all possible thresholds.
- It shows how the classifier is AWAY from the ideal, top left corner
- The worst classier would be the diagonal straight line. 50% line ie random chance.
- So, to distinguish good classifier from a bad one, Area Under Curve is used.



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC Intuition

- AUC is Area under ROC curve divided by the whole square, say 0.8 for this example.
- A poor classifier (diagonal line) will have area 0.5
- Best classifier will be (corner legs) will have area 1.
- Note that the Class labels are balanced in this example, ie number of students admitted and not admitted is roughly equal), so heights of the bell curves are same. That does not affect ROC AUC calculations though.
- Choosing threshold is a business decision.



(Ref: ROC Curves and Area Under the Curve (AUC) Explained - Data School)

ROC AUC

- ROC AUC curve can not be used to compare two models.
- As it deals with only the order of probabilities.
- So, even though model looks far better than model 2, both will have same ROC curve and thus same AUC.
- A better criteria for comparison would be log-loss.

ID	Actual	Predicted probabilities 1	Predicted probabilities 2	Predicted class
ID6	1	0.94	0.74	1
ID1	1	**0.90**	**0.69**	**1**
ID7	1	0.78	0.63	0
ID8	0	0.56	0.57	0
ID2	0	0.51	0.51	0
ID3	1	0.47	0.44	0
ID4	1	0.32	0.39	0
ID5	0	0.10	0.35	0

(Ref: Introduction to Data Science - Analytics Vidhya)

Log Loss

- As predictions are for output "1", probabilities are adjusted first.
- Then the diff (loss) is computed. Its log is taken.
- As log is negative, just to make it positive -ve sign is added.
- This procedure can be concisely put in a formula, as shown

ID	Actual	Predicted probabilities	Corrected Probabilities	Log
ID6	1	0.94	0.94	-0.0268721464
ID1	1	0.90	0.90	-0.0457574906
ID7	1	0.78	0.78	-0.1079053973
ID8	0	0.56	0.44	-0.3565473235
ID2	0	0.51	0.49	-0.30980392
ID3	1	0.47	0.47	-0.3279021421
ID4	1	0.32	0.32	-0.4948500217
ID5	0	0.10	0.90	-0.0457574906

$$H_p(q) = -\frac{1}{N}\sum_{i=1}^{N} y_i \cdot log(p(y_i)) + (1-y_i)\cdot log(1-p(y_i))$$

(Ref: Introduction to Data Science - Analytics Vidhya)

Complex Variations

Multiple Xs Logistic Regression

- How to prediction a binary response using multiple predictors?
- Can generalize the logistic function to p predictors

$$p(X) = \frac{e^{\beta_0+\beta_1 X+\ldots+\beta_p X}}{1+e^{\beta_0+\beta_1 X+\ldots+\beta_p X}} \qquad \log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

Multi Class Classification

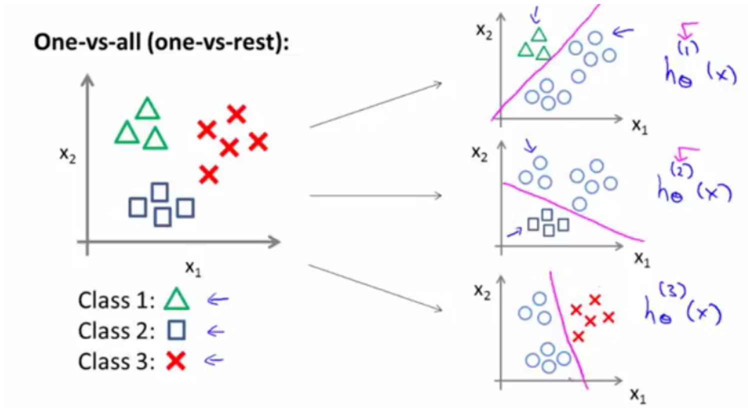
- Example: Email tagging: Work, Friends, Family, Hobby.
- 4 class problem: y = 1 or 2 or 3 or 4
- Other Example: Weather: Sunny, Cloudy, Rainy, Snow
- Target has discrete values (int)

Multi Class Logistic Regression

- How to prediction a multi-class response?
- Can generalize the logistic function to n targets?
- One vs All

One Vs All (or Rest)

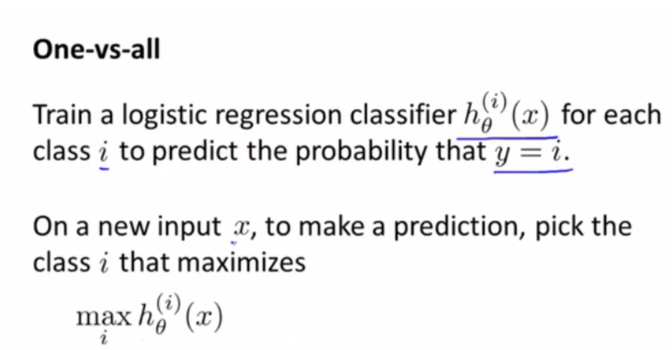
- Separate classification problems
- One class variable vs Not-that-Class
- Do this for all target variables



(Reference: Simple Linear Regression: Step-By-Step - Dan Wellisch)

One Vs All (or Rest)

- Each classifier gives its probability of its target class
- We take MAX among them



Summary

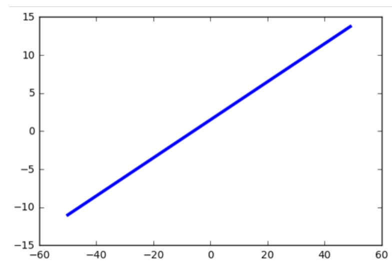
Regression vs. Classification

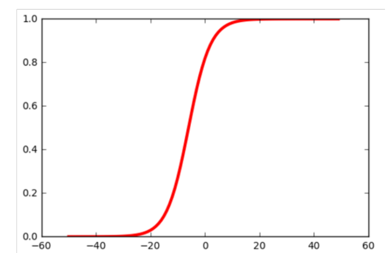
- In standard linear regression, the response is a continuous variable (float). It can take any value, large/small, positive/negative, integer/float, etc.
- In logistic regression, the response is qualitative/discrete (int, enum)
- Binary responses: Success or failure, Spam or Ham (Not Spam), etc They are bound to only categorical values such as 0,1.
- Its like converting analog signal (values, any float number) to digital signal (0 and 1s only)
- Note: Independent variables (Xs) can be of any type ie continuous or discrete, for both, Regression and Classification problems. They are differentiated based on the type of the target (y) only. If target is continuous then its a Regression problem and if the target is discrete then its a Classification problem.

Linear Model vs. Logistic Model

$$p(X) = \beta_0 + \beta_1 X$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$





Logistic Function: outputs between 0 and 1 for all values of X

Quick Check: Logistic Regression

Think About It
For the diabetes-by-age example, why does the predicted probability keep climbing toward 1 as age increases indefinitely, rather than eventually leveling off at some realistic ceiling like 0.95?

Answer
The logistic (sigmoid) curve is mathematically guaranteed to approach 1 as $z \to \infty$ and 0 as $z \to -\infty$, but never actually reach either – that's a property of the function shape, not something the model learns from data. It has no concept of a domain-specific ceiling like "max realistic risk is 95%"; it only knows the linear relationship it fit in logit space, extrapolated forever in both directions. This is exactly why extrapolating far outside the range of your training data is risky, in logistic regression just as much as linear regression.